

R22

Code No: 184BZ

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD

B. Tech II Year II Semester Examinations, June/July – 2025

OPERATING SYSTEMS

(Common to CSE, IT, CSIT, CE(SE), CSE(CS), CSE(DS), CSD)

Time: 3

Hours

Marks: 60

Max.

Note: This question paper contains two parts A and B.

i) Part-A for 10 marks, ii) Part-B for 50 marks.

- Part-A is a compulsory question which consists of ten sub-questions from all units carrying equal marks.
- Part-B consists of ten questions (numbered from 2 to 11) carrying 10 marks each. From each unit, there are two questions and the student should answer one of them. Hence, the student should answer five questions from Part-B.

PART – A

(10 Marks)

1.a) Define an operating system.

[1]

b) What are system calls in an operating system?

[1]

c) What are the main criteria used to evaluate CPU scheduling algorithms?

[1]

d) Once a deadlock is detected, what are the options for recovery?

[1]

e) What are critical regions?

[1]

f) What is Interprocess Communication (IPC)?

[1]

g) What is swapping in memory management?

[1]

h) What is the structure and role of a page table?

[1]

i) What are the different file allocation methods?

[1]

j) Define the syntax of read() system call.

[1]

PART – B

(50 Marks)

2. Describe the structure and working of a Simple Batch Operating System. How does it improve CPU utilization?

[10]

OR

3. What services does an operating system provide to users and applications? Explain each with an example.

[10]

4. Describe the Shortest Job First (SJF) scheduling algorithm. How does preemptive SJF (Shortest Remaining Time First) differ from non-preemptive SJF?

[10]

OR

5. Explain the system model used for deadlock analysis. What are the necessary conditions for a deadlock to occur?

[10]

6. Explain how semaphores can be used to solve the critical section problem. Provide a producer-consumer example using semaphores.

[10]

OR

7. Explain shared memory as an IPC mechanism. What are the benefits and challenges of using shared memory?

[10]

8. Explain the FIFO (First-In-First-Out) page replacement algorithm. What is Belady's anomaly?

[10]

OR

9. What is paging in memory management? Describe how logical addresses are translated into physical addresses in paging.

[10]

10. What is file protection in operating systems? Describe the different access control mechanisms.

[10]

OR

11. Describe the logical and physical structure of a file system. How are files organized on secondary storage?

[10]

---ooOoo---